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Unbounded strong-product ratios for the Grundy domination number

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Abstract

We prove that the ratio of the Grundy domination number of a strong product to the product of the Grundy domination numbers of its factors is unbounded over nonempty finite graphs. For each integer $m \geq 5$, we construct two graphs on the five-element subsets of an m-element set. Their Grundy domination numbers are at most m and $\binom{m}{2}$, whereas their strong product has an independent set of size $\binom{m}{5}$. Incidence matrices over two finite fields give a self-contained proof. We also determine the Grundy domination number of the graph on $\{0, \ldots, N-1\}$ with adjacency $|u - v| \equiv 2 \pmod 3$: it is $\lfloor (N+1)/3 \rfloor + 3$ for every $N \geq 6$. An explicit legal sequence in its product with the octahedron gives a counterexample on 114 vertices to the strong-product multiplicativity conjecture. Both general results have elementary proofs independent of computation. A separate Lean certificate concerns one finite example.

1. Introduction

Let G be a finite simple graph and write $N[v]$ for the closed neighbourhood of a vertex v. A sequence $(v_1, \ldots, v_k)$ of vertices of G is *legal* if every v_i dominates a vertex not dominated by its predecessors, that is,

$$N[v_i] \setminus \bigcup_{j<i} N[v_j] \neq \emptyset \qquad (1 \leq i \leq k);$$

a vertex in this difference is a *footprint* of v_i. The vertices of a legal sequence are distinct. The *Grundy domination number* $\gamma_{\mathrm{gr}}(G)$ is the length of a longest legal sequence; a longest legal sequence always dominates G. The parameter was introduced by Brešar, Gologranc, Milanič, Rall and Rizzi [7], who connected it with the domination game.

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In the strong product $G \boxtimes H$ two distinct vertices (g,h) and (g',h') are adjacent when $g' \in N_G[g]$ and $h' \in N_H[h]$; equivalently $N_{G\boxtimes H}[(g,h)] = N_G[g] \times N_H[h]$. Taking a Grundy dominating sequence of G and one of H and listing their pairs in lexicographic order gives a legal sequence of $G \boxtimes H$, so

$$\gamma_{\mathrm{gr}}(G \boxtimes H) \;\geq\; \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H) \tag{1.1}$$

for all G and H [6, Proposition 21]. In the same paper the authors conjectured that the bound is always attained.

Conjecture 1.1 ([6, Conjecture 22]). *For all graphs G and H, $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)$.*

The conjecture holds when G is a caterpillar [6] and, more generally, when G is a forest, by the work of Bell, Driscoll, Krop and Wolff [2]. A proof for all graphs was announced by Herrman and Smith [8] and later withdrawn.[1] We show that the conjecture is false and that no constant multiple of its proposed upper bound holds in general.

Our main result is the following.

Theorem 1.2. *For every $C > 0$ there are nonempty finite graphs G, H such that*

$$\gamma_{\mathrm{gr}}(G \boxtimes H) > C\,\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H).$$

Section 2 proves this using incidence matrices over two finite fields, with exact closed-neighbourhood support. The proof is independent of computation. We then give smaller counterexamples from the octahedron construction and determine the Grundy domination number of their second factors for every order.

For $N \geq 1$ let H_N be the graph on $\{0,1,\dots,N-1\}$ in which u and v are adjacent when $|u-v| \equiv 2 \pmod 3$. Let $K_{2,2,2}$ be the octahedron, the complete tripartite graph with three parts of size two.

The product construction works for every N (Proposition 4.1). We determine the Grundy domination number of its second factor without a finite computation.

Theorem 1.3. *For every $N \geq 6$,*

$$\gamma_{\mathrm{gr}}(H_N) = \left\lfloor \frac{N+1}{3} \right\rfloor + 3.$$

Moreover $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$ for every $N \geq 1$. Thus Conjecture 1.1 fails for $N = 19$ and every $N \geq 21$.

[1] The second version of arXiv:2212.04565 (12 January 2023) carries the comment "There is an error. Some cases were not considered in the proof of Theorem 1."

The proof is in Section 5. After the first selection in a suitable circulant graph, the undominated vertices form a half graph. A count at its first selected edge, or an injection at the first selected vertex of the third residue class, gives the upper bound. Explicit sequences attain it. The original finite certificate is retained separately in Appendix A.

At $N = 19$, the factors have Grundy domination numbers 2 and 9, while their product has a legal sequence of length 19. This gives a counterexample on 114 vertices. The original finite table already implied this case; no separate priority or minimality claim is made. By Proposition 25 of [6], the ratio with fixed first factor $K_{2,2,2}$ is at most 3. Theorem 1.2 varies both factors.

The original preprint is available at [3]; this submission revision and its verification files are available at [5].

2. Unbounded product ratios

To obtain unbounded ratios, we vary both factors. We use incidence matrices over different fields, following the classical intersection-congruence approach used by Alon [1, Section 3] in the study of Shannon capacity. We do not claim a new intersection-congruence method. The application here needs exact closed-neighbourhood support to bound every legal sequence. We give the construction and rank argument in full; no theorem about Shannon capacity is used as a premise.

LEMMA 2.1. *Let G be a finite graph and let M be a matrix over a field $\mathbb{F}$, with rows and columns indexed by $V(G)$. Suppose that*

$$M_{uv} \neq 0 \quad \Longleftrightarrow \quad v \in N_G[u].$$

Then $\gamma_{\mathrm{gr}}(G) \leq \operatorname{rank}_{\mathbb{F}} M$.

Proof. Let $(v_1, \ldots, v_k)$ be legal and choose a footprint w_i for each v_i. These footprints are distinct. If $i < j$, then $w_j \notin N_G[v_i]$, so $M_{v_i w_j} = 0$. Each $M_{v_i w_i}$ is nonzero. The submatrix with these rows and columns, in the stated order, is lower triangular with nonzero diagonal. Its rank is k. □

Every entry on a closed neighbourhood must be nonzero: the footprint w_i need not be v_i. A matrix that merely vanishes on nonedges would not suffice.

THEOREM 2.2. *For $m \geq 5$, let A_m and B_m have vertex set $\binom{[m]}{5}$. Distinct sets S, T are adjacent in A_m when $|S \cap T|$ is odd, and in B_m when $|S \cap T| \equiv 2 \pmod 3$. Then*

$$\gamma_{\mathrm{gr}}(A_m) \leq m, \qquad \gamma_{\mathrm{gr}}(B_m) \leq \binom{m}{2}, \qquad \gamma_{\mathrm{gr}}(A_m \boxtimes B_m) \geq \binom{m}{5}.$$

In particular,

$$\frac{\gamma_{\mathrm{gr}}(A_m \boxtimes B_m)}{\gamma_{\mathrm{gr}}(A_m)\gamma_{\mathrm{gr}}(B_m)} \geq \frac{(m-2)(m-3)(m-4)}{60m}, \tag{2.1}$$

so the ratios are unbounded.

Proof. Let U be the incidence matrix between five-element subsets of $[m]$ and elements of $[m]$. Over $\mathbb{F}_2$,

$$(UU^{\mathsf{T}})_{ST} = |S \cap T| \pmod 2.$$

Since $|S| = 5$ is odd, the nonzero entries are exactly the closed neighbourhoods of A_m. The rank is at most the number m of columns of U, so Lemma 2.1 gives the first bound.

Let V be the incidence matrix between five-element subsets and two-element subsets of $[m]$. Over $\mathbb{F}_3$,

$$(VV^{\mathsf{T}})_{ST} = \binom{|S \cap T|}{2} \pmod 3.$$

For intersection sizes $t = 0, 1, 2, 3, 4, 5$, these entries are $0, 0, 1, 0, 0, 1$. They are nonzero exactly when $S = T$ or $|S \cap T| = 2$, which gives precisely the closed neighbourhoods of B_m. Its rank is at most $\binom{m}{2}$, proving the second bound.

The product vertices

$$\mathcal{D}_m = \{(S, S) : S \in \tbinom{[m]}{5}\}$$

form an independent set. For $S \neq T$, adjacency in A_m would require intersection size 1 or 3, whereas adjacency in B_m requires size 2. Both cannot hold. Listing the vertices of $\mathcal{D}_m$ in any order gives a legal sequence, each vertex footprinting itself. This proves the lower bound $\binom{m}{5}$ for the product.

The three bounds give (2.1), whose right side tends to infinity. □

For example, $m = 13$ gives a product lower bound 1287, whereas the product of the factor bounds is $13 \cdot 78 = 1014$. This example is much larger than the octahedron example. The argument uses different fields for the two factor bounds; no matrix product over a common field is asserted. Taking m large enough proves Theorem 1.2 and answers the boundedness question left open in version 1.0.0.

3. **The graphs**

We write the vertices of the octahedron as $g_0, g_1, g_2, x_0, x_1, x_2$ (in this order, $0, \ldots, 5$), where x_t is the antipode of g_t; the three antipodal pairs are the only non-adjacent pairs (Figure 1). Thus $N[v]$ is everything except the antipode of v.

Lemma 3.1. $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$.

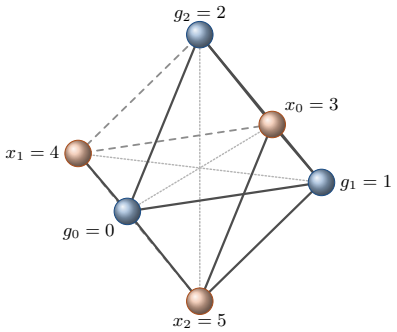


Figure 1. The octahedron $K_{2,2,2}$. Each vertex is adjacent to all others except its antipode; the dotted segments join the antipodal pairs $\{g_t, x_t\}$.

Proof. The sequence (g_0, x_0) is legal: x_0 footprints itself. Conversely, let (u, v) be legal. If v is the antipode of u, then $N[u] \cup N[v]$ is the whole vertex set; otherwise $N[u]$ misses only the antipode of u, which lies in $N[v]$. Either way nothing is left for a third vertex to footprint. □

In H_N we write $a_i = 3i$, $b_i = 3i + 1$ and $c_i = 3i + 2$. Two vertices of the same residue class are never adjacent, and for vertices of different classes,

$$a_i \sim c_j \iff j \geq i, \qquad b_i \sim a_j \iff j > i, \qquad c_i \sim b_j \iff j > i, \tag{3.1}$$

because $c_j - a_i = 3(j - i) + 2$, $a_j - b_i = 3(j - i) - 1$ and $b_j - c_i = 3(j - i) - 1$.

4. **A long legal sequence in the product**

The construction uses one property of H_N: with the colouring $v \mapsto v \bmod 3$, every edge uv with $u < v$ satisfies $u \equiv v + 1 \pmod 3$. We state it for any graph with this property.

Proposition 4.1. *Let H be a graph with vertices $v_0, \ldots, v_{N-1}$ and a map $t\colon V(H) \to \mathbb{Z}_3$ such that $t(v_i) \equiv t(v_j) + 1 \pmod 3$ for every edge $v_i v_j$ with $i < j$. Then the sequence*

$$s_j = \big(g_{t(v_j)},\, v_j\big), \qquad j = 0, 1, \ldots, N - 1,$$

is legal in $K_{2,2,2} \boxtimes H$, and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H) \geq N$. In particular $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$ for every $N \geq 1$, with $t(v) = v \bmod 3$.

Proof. We show that s_j footprints $f_j = (x_{t(v_j)+1}, v_j)$, indices of g and x taken modulo 3. First, $f_j \in N[s_j]$: the vertex $x_{t(v_j)+1}$ is not the antipode $x_{t(v_j)}$ of $g_{t(v_j)}$, so it lies in $N[g_{t(v_j)}]$, and $v_j \in N[v_j]$. Now let $i < j$. If $f_j \in N[s_i]$, then $x_{t(v_j)+1} \in N[g_{t(v_i)}]$, which means $t(v_i) \neq t(v_j) + 1$, and $v_j \in N[v_i]$, which for

$i \neq j$ means that $v_i v_j$ is an edge, hence $t(v_i) = t(v_j) + 1$. The two conclusions contradict each other, so $f_j \notin N[s_i]$. □

Figure 2 shows the sequence for H_{21}. Every column of the product contains exactly one vertex of the sequence, while a Grundy dominating sequence of the form used for (1.1) would use only $\gamma_{\mathrm{gr}}(H_{21}) = 10$ columns. The rotation of the three colours along the sequence is what lets each new vertex reach a footprint in a layer x_t that none of the earlier vertices covers in that column.

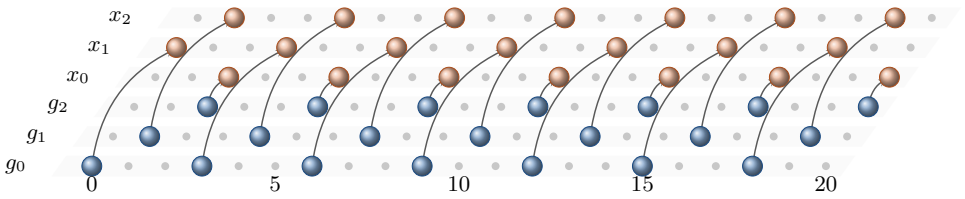


Figure 2. $K_{2,2,2} \boxtimes H_{21}$ as six stacked copies of H_{21}, one for each vertex of the octahedron. The 21 blue vertices $(g_{j \bmod 3}, j)$ form a legal sequence; the arrow from each of them ends at its footprint $(x_{(j+1) \bmod 3}, j)$.

5. The family H_N

We prove the formula for every $N \geq 6$. A legal sequence in an induced subgraph remains legal in the larger graph, with the same footprints. We use this observation for both bounds.

Lemma 5.1. *For every integer $q \geq 1$, $\gamma_{\mathrm{gr}}(H_{3q+1}) \leq q + 3$.*

Proof. Put $n = 3q + 1$. For $1 \leq d < n$, the conditions $d \equiv 2 \pmod 3$ and $n - d \equiv 2 \pmod 3$ are equivalent. Translation modulo n therefore preserves adjacency. We may translate a nonempty legal sequence so that its first vertex is 0.

This first vertex dominates 0 and all the vertices $C_k = 3k + 2$, $0 \leq k < q$. The undominated vertices are

$$X_i = 3(i + 1), \qquad Y_j = 3j + 1 \qquad (0 \leq i, j < q).$$

There are no edges within either class, and

$$X_i \sim Y_j \iff i \geq j. \tag{5.1}$$

Thus the remaining vertices induce a half graph with the perfect matching $X_i Y_i$ (Figure 3).

Follow the sequence until either a vertex C_k is selected or the selected X- and Y-vertices contain an edge for the first time. Before either event, let I, J

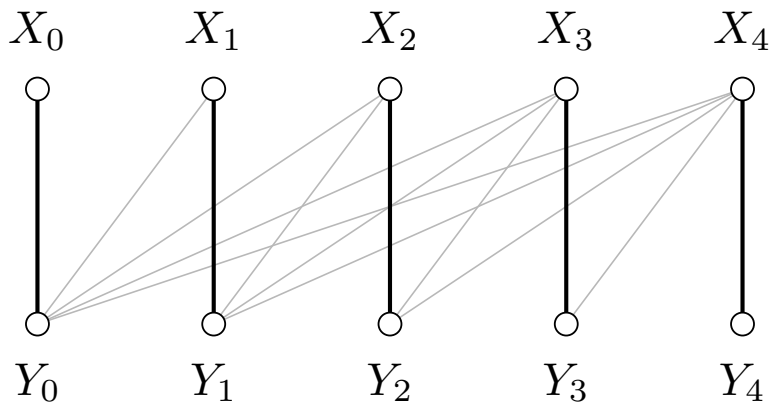


Figure 3. The undominated graph after selecting 0 in H_{3q+1}, illustrated for $q = 5$. The matching edges X_iY_i are emphasized. Before the first selected edge, every selected X-index is smaller than every selected Y-index.

be the selected indices on the two sides, and put $r = |I|$, $s = |J|$. Independence means

$$i < j \qquad (i \in I,\ j \in J). \tag{5.2}$$

Suppose first that a new X_i creates an edge. Then $J \neq \emptyset$ and $b = \min J \leq i$. All earlier X-indices are less than b, so i is the largest selected X-index. After this step, the undominated X-indices are $\{0, \dots, b-1\} \setminus I$, and the undominated Y-indices are those greater than i that are not in J. Writing $s_0 = |J \cap \{b, \dots, i\}|$, their total number is

$$b - r + (q - 1 - i) - (s - s_0).$$

The number selected since 0, plus the number still undominated, is

$$r + s + 1 + b - r + q - 1 - i - s + s_0 = q + b - i + s_0 \leq q + 1,$$

because $s_0 \leq i - b + 1$. Each later legal step must dominate a remaining vertex. The full sequence, including its initial 0, has length at most $q + 2$. If a new Y-vertex creates the first edge, use the same count after interchanging X_i and Y_{q-1-i}; this interchange preserves (5.1).

Suppose instead that C_k is selected first. Its neighbours in the other two classes are the X_i with $i < k$ and the Y_j with $j > k$. Every vertex still undominated belongs to

$$W_k = \{X_i : k \leq i < q\} \cup \{Y_j : 0 \leq j \leq k\}, \qquad |W_k| = q + 1.$$

At least $r+s$ vertices in W_k were already dominated. Indeed, map each earlier selection by

$$X_i \longmapsto \begin{cases} X_i, & i \geq k, \\ Y_i, & i < k, \end{cases} \qquad Y_j \longmapsto \begin{cases} Y_j, & j \leq k, \\ X_j, & j > k. \end{cases}$$

Each image is the selected vertex or its matching neighbour, so was already dominated. Images on each side are distinct. A collision between the sides would require an index in both I and J, contrary to (5.2). The map is injective. At most $q+1-r-s$ vertices remain undominated, and the sequence has length at most

$$1 + (r+s) + 1 + (q+1-r-s) = q+3.$$

If neither event occurs, the vertices selected after 0 form an independent set in the half graph. Its perfect matching bounds their number by q, so the full sequence has length at most $q+1$. These cases prove the lemma. □

LEMMA 5.2. *If $N \geq 6$ and $q = \lfloor (N+1)/3 \rfloor$, then $\gamma_{\mathrm{gr}}(H_N) \geq q+3$.*

Proof. For $q=2$, the sequence $(1,3,5,0,2)$ in H_6 has successive footprints $(1,5,0,2,4)$ and remains legal in H_7. For $q=3$, the sequence $(0,7,2,4,5,1)$ in H_8 has footprints $(0,7,4,6,3,1)$ and remains legal in H_9 and H_{10}.

For $q \geq 4$, it suffices to work in H_{3q-1}, which is an induced subgraph of H_N. In the notation $a_i = 3i$, $b_i = 3i+1$, $c_i = 3i+2$, take

$$a_0, a_1, \ldots, a_{q-3},\ b_{q-1},\ c_{q-3},\ c_0,\ a_{q-2},\ b_0. \tag{5.3}$$

The first $q-2$ vertices footprint themselves. The last five have successive footprints

$$b_{q-1}, \quad b_{q-2}, \quad b_{q-3}, \quad a_{q-2}, \quad a_{q-1}.$$

By (3.1), b_{q-1} is not adjacent to any earlier a_i; b_{q-2} is first dominated by c_{q-3}; and b_{q-3} is first dominated by c_0. Here $q-3 > 0$, so $c_0 \sim b_{q-3}$. Neither a_{q-2} nor a_{q-1} is adjacent to b_{q-1}, c_{q-3} or c_0. Thus a_{q-2} footprints itself, and b_0 footprints a_{q-1}. The sequence has length $q+3$. □

Proof of Theorem 1.3. Put $q = \lfloor (N+1)/3 \rfloor$. Since $N \leq 3q+1$, the graph H_N is an induced subgraph of H_{3q+1}. Lemma 5.1 gives the upper bound $q+3$, and Lemma 5.2 gives equality. The product bound is Proposition 4.1.

At $N = 19$ the formula gives $\gamma_{\mathrm{gr}}(H_{19}) = 9$. For $N \geq 21$, write $N = 3q-1$, $3q$ or $3q+1$. The respective differences $N - 2(q+3)$ are $q-7$, $q-6$ and $q-5$. Each is positive in its applicable range. Hence $N > 2\gamma_{\mathrm{gr}}(H_N)$ for $N = 19$ and every $N \geq 21$. □

6. Further questions

Question 6.1. Which graphs G satisfy Conjecture 1.1 for every H? All forests do [2], whereas the octahedron does not.

Appendix A. The finite certificate for H_{21}

LEMMA A.1. $\gamma_{\mathrm{gr}}(H_{21}) \geq 10$.

Proof. The sequence

$$\begin{aligned}(a_0, a_1, a_2, a_3, a_4, b_6, c_4, c_0, a_5, b_0)\\ = (0, 3, 6, 9, 12, 19, 14, 2, 15, 1)\end{aligned}$$

is legal and dominating. By (3.1), a_0 dominates $c_0, \dots, c_6$; each a_i with $1 \leq i \leq 4$ footprints itself and b_{i-1}; then b_6 footprints itself, c_4 footprints b_5, c_0 footprints b_4, a_5 footprints itself and b_0 footprints a_6. □

Theorem 1.3 already proves the upper bound without a computer. For an independent finite certificate we use the sets of dominated vertices. For a legal sequence $S = (v_1, \dots, v_k)$ put $D(S) = N[v_1] \cup \cdots \cup N[v_k]$. Define families $\mathcal{L}_0, \mathcal{L}_1, \dots$ of vertex sets by $\mathcal{L}_0 = \{\emptyset\}$ and

$$\mathcal{L}_{k+1} = \big\{\, D \cup N[v] \;:\; D \in \mathcal{L}_k,\ v \in V(G),\ N[v] \not\subseteq D \,\big\}.$$

LEMMA A.2. *If S is a legal sequence of length k, then $D(S) \in \mathcal{L}_k$. Hence $\gamma_{\mathrm{gr}}(G)$ is the largest k with $\mathcal{L}_k \neq \emptyset$.*

Proof. Induction on k: if $(v_1, \dots, v_{k+1})$ is legal, then $D = D(v_1, \dots, v_k) \in \mathcal{L}_k$ and $N[v_{k+1}] \not\subseteq D$, so $D \cup N[v_{k+1}] \in \mathcal{L}_{k+1}$. Conversely, every member of $\mathcal{L}_k$ is $D(S)$ for a legal sequence S of length k, by the same induction. □

The families can be large in general, but for H_{21} they are small: by (3.1), the vertices of a class dominated by the vertices of another class form an initial or a final segment of that class.[2] Their sizes are

$$|\mathcal{L}_0|, \dots, |\mathcal{L}_{10}| = 1,\ 21,\ 210,\ 1023,\ 1482,\ 1302,\ 848,\ 419,\ 150,\ 22,\ 1,$$

and $\mathcal{L}_{11} = \emptyset$; the only member of $\mathcal{L}_{10}$ is the whole vertex set. With Lemma A.2 this gives the following.

PROPOSITION A.3. $\gamma_{\mathrm{gr}}(H_{21}) \leq 10$.

We have verified Proposition A.3 in three independent ways. In Lean 4, a certificate lists the 5479 sets of the families $\mathcal{L}_0, \dots, \mathcal{L}_{10}$ as bit masks stored in binary search trees, and the kernel checks by evaluation that $\emptyset \in \mathcal{L}_0$, that

[2] For instance, the vertices c_j dominated by the chosen vertices of the class a are those with $j \geq \min\{i : a_i \text{ chosen}\}$.

$D \cup N[v] \in \mathcal{L}_{k+1}$ for every $D \in \mathcal{L}_k$ and every v with $N[v] \not\subseteq D$, and that no member of $\mathcal{L}_{10}$ has a proper extension. A formal version of Lemma A.2, proved in Lean for every graph and every such certificate, then bounds the length of every legal sequence of H_{21} by 10. The formal counterexample uses the definitions in this paper and only standard axioms; no `native_decide` is used. Independently, a C program computes the families by breadth-first search, and a Python program finds $\gamma_{\mathrm{gr}}(H_{21}) = 10$ by a memoised depth-first search over dominated sets; both use only the definition of H_{21}.

COROLLARY A.4. *$\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$, $\gamma_{\mathrm{gr}}(H_{21}) = 10$, and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_{21}) \geq 21 > 2 \cdot 10$.*

Proof. Lemma 3.1, Theorem 1.3 at $N = 21$, and Proposition 4.1 prove the result. Lemma A.1 and Proposition A.3 give the separate finite verification. □

Computations and formal verification. The repository [5] contains the Lean development, the Python and C programs, build commands and current verification results. The archived original development is [4]. Lean 4 [9] is pinned to version 4.34.1 and this development needs no external Lean library. The Lean certificate concerns only $N = 21$. Neither the all-N formula nor the unbounded-ratio theorem has been formalized. Both have complete written proofs above. Finite checks accompany them without replacing those proofs.

Use of generative AI. Claude (Anthropic) assisted with the original computations, Lean development and drafting. ChatGPT (OpenAI) assisted the exact-support rank argument and two-field construction in Section 2, using Alon's method as identified there; it also assisted the all-parameter half-graph count and explicit footprints in Section 5. OpenAI Codex assisted this revision with proof auditing, checks, organization, bibliography and typesetting. The human author is responsible for the correctness of the manuscript and its citations.

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